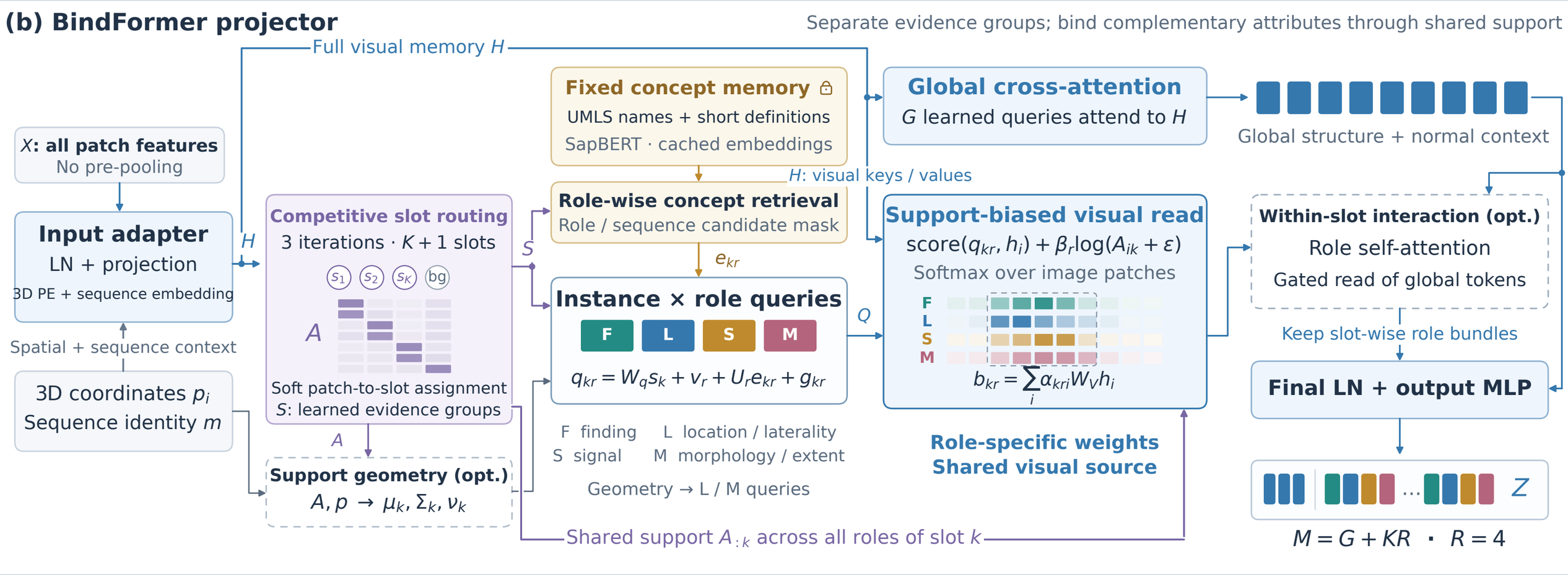

(b) BindFormer projector
Separate evidence groups; bind complementary attributes through shared support
Full visual memory H
X: all patch features
No pre-pooling
Input adapter
LN + projection
3D PE + sequence embedding
Spatial + sequence context
3D coordinates p_i
Sequence identity m
H
Competitive slot routing
3 iterations · K + 1 slots
s1 s2 sK bg
A
Soft patch-to-slot assignment
S: learned evidence groups
A
Support geometry (opt.)
A, p → μ_k, Σ_k, ν_k
S
Fixed concept memory
UMLS names + short definitions
SapBERT · cached embeddings
Role-wise concept retrieval
Role / sequence candidate mask
e_kr
Instance × role queries
F L S M
q_kr = W_q s_k + v_r + U_r e_kr + g_kr
F finding L location / laterality
S signal M morphology / extent
Geometry → L / M queries
Q
Global cross-attention
G learned queries attend to H
H: visual keys / values
Global structure + normal context
Support-biased visual read
score(q_kr, h_i) + β_r log(A_ik + ε)
Softmax over image patches
F L S M
b_kr = Σ_i α_kri W_V h_i
Role-specific weights
Shared visual source
Shared support A_:k across all roles of slot k
Within-slot interaction (opt.)
Role self-attention
Gated read of global tokens
Keep slot-wise role bundles
Final LN + output MLP
Z
M = G + KR · R = 4